

Assignment-17.1

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Batch:44

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Task 1 – Customer Data Cleaning

Use AI to generate a Python script for cleaning a customer dataset

Prompt:

Perform data preprocessing on the shopping trends dataset by handling missing values, removing duplicates, standardizing text (Location), and encoding categorical variables (Gender, Location).

Code:

```
assgn17.1 task1.py > ...
1 #Handle missing values in columns (age, income, city).
2 #Remove duplicate records.
3 #Standardize text values (e.g., city names in lowercase).
4 #Encode categorical variables (gender, city).
5 import pandas as pd
6 data = pd.read_csv("C:\\Users\\sushm\\Downloads\\shopping_trends.csv")
7 # Optional: clean column names (recommended)
8 data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")
9 # Handle missing values
10 data = data.dropna(subset=['age', 'purchase_amount_(usd)', 'location'])
11 # Remove duplicates
12 data = data.drop_duplicates()
13 # Standardize text
14 data['location'] = data['location'].str.lower().str.strip()
15 # Encode categorical variables
16 data = pd.get_dummies(data, columns=['gender', 'location'], drop_first=True)
17 print(data.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assgn17.1 task1.py"
```

	customer_id	age	item_purchased	...	location_west	virginia	location_wisconsin	location_wyoming
0	1	55	Blouse	...	False	False	False	False
1	2	19	Sweater	...	False	False	False	False
2	3	50	Jeans	...	False	False	False	False
3	4	21	Sandals	...	False	False	False	False
4	5	45	Blouse	...	False	False	False	False

[5 rows x 67 columns]

Observation:

- Missing values in key columns were removed to improve data quality and avoid errors during analysis.
- Duplicate records were identified and deleted, ensuring no repeated entries affect results.
- The **Location** column was standardized to lowercase and trimmed, maintaining consistency in text data.
- Categorical variables like **Gender** and **Location** were converted into numerical format using one-hot encoding.
- After preprocessing, the dataset became clean, consistent, and suitable for machine learning models and further analysis.

Task 2 – Sales Data Preprocessing

Preprocess sales transaction data using AI assistance.

Prompt:

Perform data preprocessing on the sales dataset by converting the date column into datetime format, extracting new features (month, year, day of week), normalizing the sales amount, and detecting and removing outliers.

Code:

```
assgn17.1 task2.py >...
1 #Convert date columns into datetime format.
2 # Create new features (month, year, day of week).
3 #Normalize sales amount column.
4 #Detect and handle outliers in dataset.
5 import pandas as pd
6 data = pd.read_csv("C:\\Users\\sushm\\Downloads\\sales_data.csv")
7 data.columns = data.columns.str.strip().str.lower()
8 print(data.columns) # check once
9 # Convert date column
10 data['sale_date'] = pd.to_datetime(data['sale_date'])
11 # Create features
12 data['sale_month'] = data['sale_date'].dt.month
13 data['sale_year'] = data['sale_date'].dt.year
14 data['sale_dayofweek'] = data['sale_date'].dt.dayofweek
15 # Normalize
16 data['sales_amount_normalized'] = (
17     (data['sales_amount'] - data['sales_amount'].min()) /
18     (data['sales_amount'].max() - data['sales_amount'].min()))
19 )
20 Q1 = data['sales_amount'].quantile(0.25)
21 Q3 = data['sales_amount'].quantile(0.75)
22 IQR = Q3 - Q1
23 lb = Q1 - 1.5 * IQR
24 ub = Q3 + 1.5 * IQR
25 data = data[(data['sales_amount'] >= lb) & (data['sales_amount'] <= ub)]
26 print(data.head())
```

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```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe
xe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assgn17.1 task2.py"
product_id sale_date sales_rep region ... sale_month sale_year sale_dayofweek sales_amount_normalized
0 1052 2023-02-03 Bob North ... 2 2023 4 0.500950
1 1093 2023-04-21 Bob West ... 4 2023 4 0.433202
2 1015 2023-09-21 David South ... 9 2023 3 0.458201
3 1072 2023-08-24 Bob South ... 8 2023 3 0.209105
4 1061 2023-03-24 Charlie East ... 3 2023 4 0.369108
```

Observation:

- The **sale_date** column was successfully converted into datetime format, enabling time-based analysis.
- New features such as **month, year, and day of week** were extracted, enhancing the dataset with useful temporal information.
- The **sales_amount** column was normalized to a range between 0 and 1, making it suitable for machine learning models.

- Outliers in the sales data were detected using the **IQR method** and removed to improve data consistency.
- After preprocessing, the dataset became structured, clean, and more suitable for analysis and predictive modeling.

Task 3 – Healthcare Data Preparation

Clean and preprocess a healthcare dataset using AI scripts.

Prompt:

Perform preprocessing on health dataset including handling missing values, encoding categorical features, scaling numerical features, and splitting into train and test sets.

Code:

```

+ assign17.1 task3.py > ...
1  #Handle missing values in blood_pressure, cholesterol.
2  #Encode categorical features
3  #Scale numerical features (min-max or standard scaling).
4  #Split dataset into training and testing sets.
5  import pandas as pd
6
7  # Load dataset
8  data = pd.read_csv(r"C:\Users\sushm\Downloads\dirty_v3_path.csv")
9
10 # Clean column names
11 data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")
12
13 # Handle missing values
14 data = data.dropna(subset=['blood_pressure', 'cholesterol'])
15
16 # Encode categorical features
17 data = pd.get_dummies(data, columns=[
18     'gender', 'smoking', 'alcohol', 'family_history'
19 ], drop_first=True)
20
21 # Split dataset (using medical_condition as target)
22 X = data.drop('medical_condition', axis=1)
23 y = data['medical_condition']
24
25 # Train-test split
26 from sklearn.model_selection import train_test_split
27 X_train, X_test, y_train, y_test = train_test_split(
28     X, y, test_size=0.1, random_state=42)
29
30 # Train the model
31 model = LogisticRegression()
32 model.fit(X_train, y_train)
33
34 # Evaluate the model
35 score = model.score(X_test, y_test)
36 print(f"Model Score: {score}")
37
38 # Save the model
39 joblib.dump(model, 'model.pkl')
40
41 # Load the model
42 model = joblib.load('model.pkl')
43
44 # Predict on new data
45 new_data = pd.read_csv('new_data.csv')
46 new_data = new_data.dropna(subset=['blood_pressure', 'cholesterol'])
47 new_data = pd.get_dummies(new_data, columns=[
48     'gender', 'smoking', 'alcohol', 'family_history'
49 ], drop_first=True)
50 new_X = new_data.drop('medical_condition', axis=1)
51 new_y = new_data['medical_condition']
52
53 # Predict
54 new_predictions = model.predict(new_X)
55
56 # Print predictions
57 print(new_predictions)
58
59 # End of script
60
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/D
• PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/D
esktop/Ai-Assist/assign17.1 task3.py"
X_train: (20400, 19)
X_test: (5100, 19)
y_train: (20400,)
y_test: (5100,)
○ PS C:\Myfiles\Attachments\Desktop\Ai-Assist>

```

```

25 # Train-test split
26 from sklearn.model_selection import train_test_split
27 X_train, X_test, y_train, y_test = train_test_split(
28     X, y, test_size=0.2, random_state=42
29 )
30
31 # Scale numerical features
32 from sklearn.preprocessing import MinMaxScaler
33
34 scaler = MinMaxScaler()
35 num_cols = ['age', 'blood_pressure', 'cholesterol']
36
37 X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
38 X_test[num_cols] = scaler.transform(X_test[num_cols])
39
40 # Output
41 print("X_train:", X_train.shape)
42 print("X_test:", X_test.shape)
43 print("y_train:", y_train.shape)
44 print("y_test:", y_test.shape)

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Att
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Att
esktop/Ai-Assist/assgn17.1 task3.py"
X_train: (20400, 19)
X_test: (5100, 19)
y_train: (20400,)
y_test: (5100,)
PS C:\Myfiles\Attachments\Desktop\Ai-Assist>

```

Observation:

Missing values were removed, categorical features were encoded, numerical features were scaled, and the dataset was split into training and testing sets for model building.

Task 4 – Employee Salary Data Preprocessing

Clean and preprocess an employee salary dataset using AI scripts.

Prompt:

Perform data preprocessing on the employee salary dataset by handling missing values, detecting and removing salary outliers, standardizing gender values, and encoding categorical features.

Code:

```

1 #Handle missing values
2 #Detect and remove salary outliers.
3 #Standardize department names (lowercase).
4 #Apply label encoding to job location, department.
5 import pandas as pd
6
7 # Load dataset (make sure it's CSV, not ipynb)
8 data = pd.read_csv(r"C:\Users\sushm\Downloads\Employee_Salary_Dataset.csv")
9
10 # Clean column names
11 data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")
12
13 # Handle missing values
14 data = data.dropna(subset=['salary'])
15
16 # Remove salary outliers (IQR)
17 Q1 = data['salary'].quantile(0.25)
18 Q3 = data['salary'].quantile(0.75)
19 IQR = Q3 - Q1
20
21 lb = Q1 - 1.5 * IQR
22 ub = Q3 + 1.5 * IQR
23
24 data = data[(data['salary'] >= lb) & (data['salary'] <= ub)]
25
26 # Standardize gender
27 data['gender'] = data['gender'].str.lower().str.strip()

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assign17.1 task4.py"

	id	experience_years	age	gender	salary
0	1	5	28	0	250000
1	2	1	21	1	50000
2	3	3	23	0	170000
3	4	2	22	1	25000
4	5	1	17	1	10000

```

18 Q3 = data['salary'].quantile(0.75)
19 IQR = Q3 - Q1
20
21 lb = Q1 - 1.5 * IQR
22 ub = Q3 + 1.5 * IQR
23
24 data = data[(data['salary'] >= lb) & (data['salary'] <= ub)]
25
26 # Standardize gender
27 data['gender'] = data['gender'].str.lower().str.strip()
28
29 # Encode gender
30 from sklearn.preprocessing import LabelEncoder
31 le = LabelEncoder()
32 data['gender'] = le.fit_transform(data['gender'])
33
34 # Output
35 print(data.head())

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assign17.1 task4.py"

	id	experience_years	age	gender	salary
0	1	5	28	0	250000
1	2	1	21	1	50000
2	3	3	23	0	170000
3	4	2	22	1	25000
4	5	1	17	1	10000

Observation:

- Missing values in the **salary** column were removed to ensure data quality.
- Outliers in salary were detected using the **IQR method** and eliminated to avoid skewed analysis.
- The **gender** column was standardized to lowercase for consistency.
- Categorical data (gender) was encoded into numerical format using label encoding.
- The dataset is now clean, consistent, and ready for further analysis or machine learning tasks.